

LAB 4 SAMPLE OUTPUT

Note that the time to run these algorithms on your computer WILL be **different** then what is in the sample output. However, you may see similarities in comparing the algorithms.

Runtime can vary on different computers because of differences in the

- CPU (clock speed, number of cores, CPU architecture),
- cache size & memory hierarchy,
- amount of RAM (main memory),
- speed of RAM,
- whether other programs are using main memory at the same time,
- storage type,
- operating system, and
- background processes running at the same time as the program.

Sample Output 1 – Running With `movies_short.txt`

- User input is highlighted in **yellow**.
- You need to run your program with this output first to make sure your sort algorithms are implemented correctly.
- Make sure to scramble the arrays and print again before running each sort function.

```
Filename (movies.txt): movies_short.txt
Array size doubled from 20 to 40.
Loaded 37 movies from 'movies_short.txt'.
```

```
===== MENU =====
```

```
1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 1
```

```
Halloween (1978)
Friday the 13th (1980)
A Nightmare on Elm Street (1984)
Scream (1996)
The Exorcist (1973)
The Texas Chain Saw Massacre (1974)
Texas Chainsaw 3D (2013)
Leatherface (2017)
Texas Chainsaw Massacre (2022)
The Conjuring (2013)
Annabelle (2014)
The Nun (2018)
The Curse of La Llorona (2019)
It (2017)
Psycho (1960)
The Shining (1980)
Doctor Sleep (2019)
Poltergeist (1982)
Child's Play (1988)
The Ring (2002)
Paranormal Activity (2007)
```

Saw (2004)
Get Out (2017)
Us (2019)
Nope (2022)
Hereditary (2018)
Midsommar (2019)
The Babadook (2014)
Insidious (2010)
The Cabin in the Woods (2012)
Trick 'r Treat (2007)
Hocus Pocus (1993)
Casper (1995)
Beetlejuice (1988)
Coraline (2009)
Monster House (2006)
The Addams Family (1991)

===== MENU =====

- 1) Print movies
- 2) Bubble sort (A->Z)
- 3) Selection sort (A->Z)
- 4) Insertion sort (A->Z)
- 5) Scramble array (randomize order)
- 6) Quit

Choose an option (1-6): 2

I am going to bubble sort the array of 37 movies
Bubble sort completed in 0 ms.

===== MENU =====

- 1) Print movies
- 2) Bubble sort (A->Z)
- 3) Selection sort (A->Z)
- 4) Insertion sort (A->Z)
- 5) Scramble array (randomize order)
- 6) Quit

Choose an option (1-6): 1

A Nightmare on Elm Street (1984)
Annabelle (2014)
Beetlejuice (1988)
Casper (1995)
Child's Play (1988)
Coraline (2009)
Doctor Sleep (2019)
Friday the 13th (1980)
Get Out (2017)
Halloween (1978)
Hereditary (2018)
Hocus Pocus (1993)
Insidious (2010)
It (2017)
Leatherface (2017)
Midsommar (2019)
Monster House (2006)
Nope (2022)
Paranormal Activity (2007)
Poltergeist (1982)
Psycho (1960)
Saw (2004)
Scream (1996)
Texas Chainsaw 3D (2013)
Texas Chainsaw Massacre (2022)

The Addams Family (1991)
The Babadook (2014)
The Cabin in the Woods (2012)
The Conjuring (2013)
The Curse of La Llorona (2019)
The Exorcist (1973)
The Nun (2018)
The Ring (2002)
The Shining (1980)
The Texas Chain Saw Massacre (1974)
Trick 'r Treat (2007)
Us (2019)

===== MENU =====

1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 5

Array scrambled.

===== MENU =====

1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 1

The Exorcist (1973)
Nope (2022)
Texas Chainsaw 3D (2013)
Midsommar (2019)
Insidious (2010)
Get Out (2017)
Monster House (2006)
Saw (2004)
The Curse of La Llorona (2019)
Hocus Pocus (1993)
It (2017)
The Shining (1980)
Friday the 13th (1980)
The Conjuring (2013)
Paranormal Activity (2007)
The Texas Chain Saw Massacre (1974)
Casper (1995)
Child's Play (1988)
Leatherface (2017)
Texas Chainsaw Massacre (2022)
The Addams Family (1991)
Doctor Sleep (2019)
Scream (1996)
Hereditary (2018)
Halloween (1978)
Coraline (2009)
The Babadook (2014)
The Cabin in the Woods (2012)
Us (2019)
The Nun (2018)

A Nightmare on Elm Street (1984)
The Ring (2002)
Beetlejuice (1988)
Trick 'r Treat (2007)
Psycho (1960)
Annabelle (2014)
Poltergeist (1982)

===== MENU =====

- 1) Print movies
- 2) Bubble sort (A->Z)
- 3) Selection sort (A->Z)
- 4) Insertion sort (A->Z)
- 5) Scramble array (randomize order)
- 6) Quit

Choose an option (1-6): 3

I am going to selection sort the array of 37 movies
Selection sort completed in 0 ms.

===== MENU =====

- 1) Print movies
- 2) Bubble sort (A->Z)
- 3) Selection sort (A->Z)
- 4) Insertion sort (A->Z)
- 5) Scramble array (randomize order)
- 6) Quit

Choose an option (1-6): 1

A Nightmare on Elm Street (1984)
Annabelle (2014)
Beetlejuice (1988)
Casper (1995)
Child's Play (1988)
Coraline (2009)
Doctor Sleep (2019)
Friday the 13th (1980)
Get Out (2017)
Halloween (1978)
Hereditary (2018)
Hocus Pocus (1993)
Insidious (2010)
It (2017)
Leatherface (2017)
Midsommar (2019)
Monster House (2006)
Nope (2022)
Paranormal Activity (2007)
Poltergeist (1982)
Psycho (1960)
Saw (2004)
Scream (1996)
Texas Chainsaw 3D (2013)
Texas Chainsaw Massacre (2022)
The Addams Family (1991)
The Babadook (2014)
The Cabin in the Woods (2012)
The Conjuring (2013)
The Curse of La Llorona (2019)
The Exorcist (1973)
The Nun (2018)
The Ring (2002)
The Shining (1980)

The Texas Chain Saw Massacre (1974)
Trick 'r Treat (2007)
Us (2019)

===== MENU =====

1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 5

Array scrambled.

===== MENU =====

1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 4

I am going to insertion sort the array of 37 movies
Insertion sort completed in 0 ms.

===== MENU =====

1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 1

A Nightmare on Elm Street (1984)
Annabelle (2014)
Beetlejuice (1988)
Casper (1995)
Child's Play (1988)
Coraline (2009)
Doctor Sleep (2019)
Friday the 13th (1980)
Get Out (2017)
Halloween (1978)
Hereditary (2018)
Hocus Pocus (1993)
Insidious (2010)
It (2017)
Leatherface (2017)
Midsommar (2019)
Monster House (2006)
Nope (2022)
Paranormal Activity (2007)
Poltergeist (1982)
Psycho (1960)
Saw (2004)
Scream (1996)
Texas Chainsaw 3D (2013)
Texas Chainsaw Massacre (2022)
The Addams Family (1991)
The Babadook (2014)

The Cabin in the Woods (2012)
The Conjuring (2013)
The Curse of La Llorona (2019)
The Exorcist (1973)
The Nun (2018)
The Ring (2002)
The Shining (1980)
The Texas Chain Saw Massacre (1974)
Trick 'r Treat (2007)
Us (2019)

===== MENU =====

1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 5

Array scrambled.

===== MENU =====

1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 6

Goodbye!

Sample Output 2 – Running With movies.txt

- User input is highlighted in yellow.
- Run your program with this sample output. You will get a different number of ms for each algorithm, but bubble sort should take the longest.
- Notice I never print the movies because I already know my algorithms work.
- Also, notice I always scramble array after calling each sort algorithm, however at the end I call bubble sort twice in a row to show that if you bubble sort an already sorted array, it runs very quickly because I implemented the more efficient version of bubble sort. Yours should do this too!

Filename (movies.txt): movies.txt
Array size doubled from 20 to 40.
Array size doubled from 40 to 80.
Array size doubled from 80 to 160.
Array size doubled from 160 to 320.
Array size doubled from 320 to 640.
Array size doubled from 640 to 1280.
Array size doubled from 1280 to 2560.
Array size doubled from 2560 to 5120.
Loaded 4661 movies from 'movies.txt'.

===== MENU =====

1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)

6) Quit

Choose an option (1-6): 2

I am going to bubble sort the array of 4661 movies
Bubble sort completed in 194 ms.

===== MENU =====

- 1) Print movies
- 2) Bubble sort (A->Z)
- 3) Selection sort (A->Z)
- 4) Insertion sort (A->Z)
- 5) Scramble array (randomize order)
- 6) Quit

Choose an option (1-6): 5

Array scrambled.

===== MENU =====

- 1) Print movies
- 2) Bubble sort (A->Z)
- 3) Selection sort (A->Z)
- 4) Insertion sort (A->Z)
- 5) Scramble array (randomize order)
- 6) Quit

Choose an option (1-6): 3

I am going to selection sort the array of 4661 movies
Selection sort completed in 69 ms.

===== MENU =====

- 1) Print movies
- 2) Bubble sort (A->Z)
- 3) Selection sort (A->Z)
- 4) Insertion sort (A->Z)
- 5) Scramble array (randomize order)
- 6) Quit

Choose an option (1-6): 5

Array scrambled.

===== MENU =====

- 1) Print movies
- 2) Bubble sort (A->Z)
- 3) Selection sort (A->Z)
- 4) Insertion sort (A->Z)
- 5) Scramble array (randomize order)
- 6) Quit

Choose an option (1-6): 4

I am going to insertion sort the array of 4661 movies
Insertion sort completed in 78 ms.

===== MENU =====

- 1) Print movies
- 2) Bubble sort (A->Z)
- 3) Selection sort (A->Z)
- 4) Insertion sort (A->Z)
- 5) Scramble array (randomize order)
- 6) Quit

Choose an option (1-6): 5

Array scrambled.

```

===== MENU =====
1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 2

I am going to bubble sort the array of 4661 movies
Bubble sort completed in 130 ms.

===== MENU =====
1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 2

I am going to bubble sort the array of 4661 movies
Bubble sort completed in 0 ms.

===== MENU =====
1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 6

Goodbye!

```

SAMPLE OUTPUT 3 – JUST FOR FUN – MOVIES_VERY_LONG.TXT

- User input is highlighted in yellow.
- I ended up running the program on the long movie list just so you all can see how long it takes (on my computer).

```

Filename (movies.txt): movies_very_long.txt

Array size doubled from 20 to 40.
Array size doubled from 40 to 80.
Array size doubled from 80 to 160.
Array size doubled from 160 to 320.
Array size doubled from 320 to 640.
Array size doubled from 640 to 1280.
Array size doubled from 1280 to 2560.
Array size doubled from 2560 to 5120.
Array size doubled from 5120 to 10240.
Array size doubled from 10240 to 20480.
Array size doubled from 20480 to 40960.
Array size doubled from 40960 to 81920.
Array size doubled from 81920 to 163840.
Array size doubled from 163840 to 327680.
Loaded 233547 movies from 'movies_very_long.txt'.

===== MENU =====

```



```
1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 2
```

I am going to bubble sort the array of 233547 movies
Bubble sort completed in 631353 ms.

```
===== MENU =====
1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 5
Array scrambled.
```

```
===== MENU =====
1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 3
```

I am going to selection sort the array of 233547 movies
Selection sort completed in 564248 ms.

```
===== MENU =====
1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 5

Array scrambled.
```

```
===== MENU =====
1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
Choose an option (1-6): 4
```

I am going to insertion sort the array of 233547 movies
Insertion sort completed in 582067 ms.

```
===== MENU =====
1) Print movies
2) Bubble sort (A->Z)
3) Selection sort (A->Z)
4) Insertion sort (A->Z)
5) Scramble array (randomize order)
6) Quit
```

Choose an option (1-6): 6

Goodbye!